

Wavelet-based enhancement network for low-light image[☆]XiaoPeng Hu^a, Kang Liu^a , Xiangchen Yin^b, Xin Gao^a , Pingsheng Jiang^a, Xu Qian^a^a School of Artificial Intelligence, China University of Mining and Technology-Beijing, Beijing 100083, China^b University of Science and Technology of China, Hefei 230026, China

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ABSTRACT

Low-light images are key challenges for high-level vision tasks, often leading to failures in intelligent systems. To achieve more robust low-light enhancement and gain improvement for downstream segmentation task, in this paper we propose a wavelet-based enhancement network (WENet) that combines convolution layer and Transformer block. The wavelet transform separates different frequency components from the multi-scale transformation of the signal. We propose a wavelet calibrate layer (WCL), which converts the feature to the wavelet domain and distributes it to the corresponding area through multiple calibration filters, and restores details of the image. Recognizing that noise amplification occurs concurrently with wavelet learning, we build a contrast adjustment layer (CAL), which refines the contrast primarily through shift operations. WENet has achieved superior performance on the LOL, LOLv2 and MIT-Adobe FiveK datasets for enjoyable visual experience, reaching 22.34 and 0.814 on PSNR and SSIM respectively. We trained WENet and segmentation model by end-to-end in the dark scene of ACDC dataset and achieved advanced effect, which is robust for low-light scenes.

1. Introduction

As deep learning has surged in popularity, computer vision has seen explosive growth. Despite this, the majority of vision models are developed and evaluated under normal conditions using standard datasets. In low-light environments (such as at night, uneven lighting, etc.) images always have low contrast and visibility occur, and the darker the area in the image, the more information is lost and harder to recover, which causes many computer vision systems to fail [1,2]. Low-light enhancement not only improves visual pleasure, but also provides benefits for downstream high-level visual tasks such as object detection [3–5], semantic segmentation [6,7]. High-quality images are necessary for the robustness of the model, so low-light enhancement can potentially be applied in many fields [8,9]. Fig. 1 shows the improvement of low-light enhancement in a semantic segmentation task. The enhanced images not only improve nighttime segmentation effects but also accurately predict the presence and location of obscured objects such as the trucks indicated in the image.

Traditional image enhancement is mainly divided into histogram-based methods [10–12] and Retinex-based methods [13–15]. On the one hand, many researchers initially balanced the dynamic range of images by statistical distribution of pixel intensity and color. Histogram

equalization and its variants [16,17] are simple and effective in enhancing images especially contrast, but the recovery of image details is limited. On the other hand, the core idea of Retinex theory [18] is to remove the influence of light in the image and retain the unique properties of the object in the image. This theory can be described as

$$S(x, y) = I(x, y) \times R(x, y) \quad (1)$$

where $S(x, y)$ is the image as input, $I(x, y)$ is the illumination component, and $R(x, y)$ is the reflection component. The image can be broken down into illumination components and reflection components, which represent the illuminance distribution and the properties of the object, respectively. The method based on Retinex achieves good results through component decomposition, but the theory is not applicable in all conditions due to the lack of strong constraints. Decomposition often results in unnatural images and an unpleasant experience for the user. Some methods [19,20] solve this problem by introducing some additional priors or regularizers, but these manually designed constraints are difficult to apply to all scenarios. These methods rely on mathematical and statistical theory to achieve good results. However, due to depend on manually setting parameters, they do not generalize about the loss of image information and the noise that occurs. In addition, these methods use complex mathematical calculations to obtain

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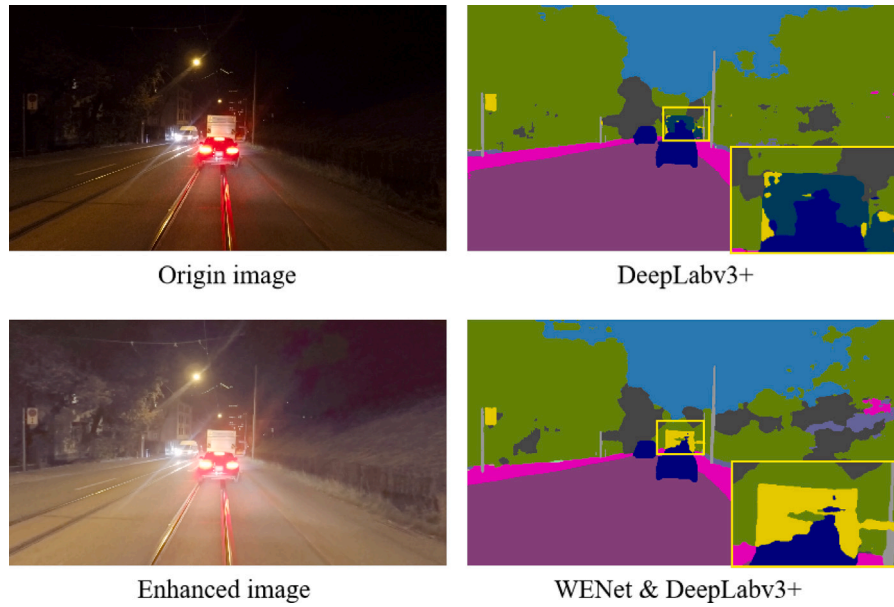


Fig. 1. Example of low-light enhancement for segmentation task, it is shown that the enhancement image can improve the performance of semantic segmentation. The image compares the original low-light image with the enhanced image in WENet for night semantic segmentation effect, which can effectively predict the blocked object after enhancement.

enhanced images, they put great pressure on the real-time performance of the visual system.

With the popularity of deep learning in recent years, the powerful feature learning capability of convolutional neural networks (CNN) brings new development potential to low-light enhancement (LLIE) [21–24], and deep learning-based models have better accuracy and speed. Convolutional Neural Networks (CNNs) have adeptly embraced the challenge of low-light image enhancement, reformulating the task as an image-to-image translation problem. This approach transcends traditional enhancement methods by eschewing reliance on prior knowledge or physical models, thus enabling a more flexible and dynamic image enhancement strategy. The models developed for this purpose can be categorized into three distinct paradigms: supervised learning, unsupervised learning, and zero-shot learning. A supervised learning-based approach [25,26] uses pairs of low-light and normal images to learn the enhanced mapping, directly resulting in the enhanced image. In addition, some models [27] will be combined with Retinex theory to separate the illumination Component and reflection Component, and enhance and adjust them respectively. Compared with the traditional Retinex-based method, the prior parameters can be more accurate through the network. However, most of the existing datasets are made by adjusting the exposure, and the model may be overfitting. For this purpose, unsupervised and semi-supervised methods have emerged, which do not require pairs of training data or even learn only from test images. EnlightenGAN [28] are proposed to train against adversarial learning by using U-Net as a generator. Zero-DCE [29] uses low-light enhancement as a task for image-specific curve estimation using deep networks. Ma et al. [30] proposed a novel self-calibrating lighting (SCI) framework and established weight-sharing lighting learning. IAT [31] explores real-time dark light enhancement and exposure correction. These methods can significantly improve image brightness and contrast, resulting in a nice visual experience. However, they are not obvious in suppressing noise in low light regions, the recovery of image information is limited, and the gain of downstream visual tasks is very limited.

A lot of work [32,33] is implemented in the spatial domain. Sun et al. [34] propose a multi-resolution network to remove moire patterns at different frequencies and scales. Liu et al. [35] propose a novel method based on wavelet and double branch neural network for image moire removal. We are convinced that processing low-light image

enhancement in the wavelet domain offers distinct advantages, particularly in terms of more effectively filtering out high-frequency noise and recovering lost image information. Existing wavelet-based methods have been used in several domains, such as face super-resolution [36], deblurring [37], and classification [38]. By doing so, we aim to achieve significant enhancements in performance for downstream semantic segmentation tasks, which are critical for robust computer vision systems. Furthermore, we are committed to designing our enhancement model with lightweight processing in mind, ensuring it meets the stringent computational demands of real-world computer vision applications.

In this paper, we introduce the Wavelet-based Enhancement Network (WENet), a novel framework that integrates CNNs and Transformers to enhance low-light images within the wavelet domain. Our approach commences with preliminary feature processing facilitated by a Transformer block, leveraging the wavelet domain inherent ability to effectively filter out noise. Wavelet domain is much better for networks to filter out noise, we propose a wavelet calibrate layer (WCL) that converts features to the wavelet domain and then distributes them to the corresponding region calibration through multiple filters, which brings great semantic learning capability while preserving translation invariance. Then we convert the features back to the spatial domain by inverse wavelet transform. In addition, we build a contrast adjustment layer (CAL) to further enhance the feature for improving contrast, which is only made up of convolutional layers. Extensive qualitative and quantitative experiments demonstrate that WENet achieved advanced performance on the LOL [27] and LOLv2 datasets. We verify the effect of WENet in the segmentation task in the dark scene of ACDC [39] dataset, and combine WENet with the segmentation model for training and fine-tuning, achieving good segmentation effect.

Our contribution can be summarized as follows:

- Leveraging the power of wavelet transform, we propose the Wavelet-based Enhancement Network (WENet), a novel architecture that synergizes convolutional layers with Transformer blocks. This integration is designed to efficiently filter out noise and recover valuable information within the wavelet domain, thereby enhancing the quality of low-light images with greater precision and effectiveness.
- We construct a wavelet calibrate layer (WCL) and correct the features in the wavelet domain to recover details of the image. In addition, we construct a contrast adjustment layer (CAL) which

is completely composed of convolutional layers to enhance the features again and improve the contrast of the image.

- We conduct extensive qualitative and quantitative experiments, our WENet effectively filters noise and improves the performance of high-level visual task.

2. Related work

2.1. Low-light image enhancement

Traditional image enhancement is divided into histogram-based method and Retinex-based method. Reza et al. [40] propose that local equalization with clipping constraints shows good performance, but some scenarios have serious semantic loss. Ibrahim et al. [11] proposed a dynamic histogram equalization of brightness preservation (BPDHE). First, the method uses a one-dimensional Gaussian filter to input the histogram, and then performs local partitioning according to the maximum value. Guo et al. [13] proposed a low-light image enhancement (LIME) method. First, by looking for the maximum value in the R, G, and B channels, the initial light map is refined by adding a structure as the final enhanced light map. Wang et al. [19] propose a bright-pass filter that constrains the range of reflectivity components to estimate illumination and reflectivity components. Kim et al. [41] attempt to estimate the illumination component by taking the maximum value in multiple diffusion spaces at each pixel position, while noise does not expand.

With the popularity of convolutional neural networks (CNNs), low-light enhancement has seen explosive growth [42,43]. For the first time since LLNet [44] used autoencoders to de-noise images, CNN was used for low light enhancement. Chen et al. proposed RetinexNet [27], which for the first time combined Retinex theory with deep learning for image enhancement. RetinexNet decomposes the image into a Reflection Component and an Illumination Component by decomposing the network. Then the component is reconstructed by the enhanced network and the adjusted network respectively, and finally the enhanced image is obtained by Retinex theory. In addition, they propose the LOL dataset as a baseline dataset for low light enhancement. Zhang et al. [25] proposed Kind, which is also combined with Retinex theory. Unlike RetinexNet, it does not require low-light-normal light image pairs, only a bunch of images under different exposure conditions to train the network, saving on the cost of the dataset. In addition, Kind can freely adjust the scale of enhancement through parameters. Lv et al. [26] proposed a multi-branch network MBLEN, which can simultaneously achieve noise reduction and low light enhancement. For unknown visual scenes where existing low-light enhancement models are not robust, Ma et al. [30] propose a novel self-calibrated illumination (SCI) framework to achieve fast and robust image enhancement. They built a light learning with weight sharing and built self-calibration modules to achieve convergence at each stage. SCI uses unsupervised loss to improve adaptation to general scenarios. Downstream tasks such as dark face detection and semantic segmentation at night show the potential utility of SCI. Jiang et al. [45] proposed Real-low to Real-normal Network (R2 RNet) for low-light image enhancement, which uses spatial information to improve contrast and frequency information to preserve details. Guo et al. [29] propose Zero-DCE, a method that transforms low-light enhancement into the task of image-specific curve estimation, constrained by a set of carefully designed mathematical functions. Zhou et al. [46] proposed a low-light enhancement network named GLARE based on codebook retrieval with latent features, providing a new research perspective. Xu et al. [47] proposed an Unbalanced Points-guided multi-scale Transformer-based conditional normalizing Flow (UPT-Flow) and designed cross-coupling conditional affine layers. Ma et al. proposed [48] a low-light image enhancement method based on Retinex and integrated image features in channel and spatial attention mechanisms to improve the ability of network. Wan et al.

proposed [49] the Patch-based Simplified Conditional Diffusion Model (PSC Diffusion) for low-light enhancement. They adopted SimpleGate and parameter-free attention (SimPF) blocks to predict noises of the diffusion model. Wan et al. proposed [50] SFDiff, which integrates Fourier frequency information into the diffusion process. Meanwhile, they adopted the spatial frequency fusion (SFF) block to fully integrate and interact information through spatial and frequency domains. Most of the current low light enhancement methods have complex parameters, which is not conducive to real-time operation. In addition, these models enhance the brightness of the image too much, and the gain for detail recovery and downstream visual tasks is limited.

2.2. Wavelet-based model

After wavelet transform, the image is decomposed into the combination of different wavelet coefficients. And after each wavelet transform, the image is divided into a quarter of the subband region, which contains the content region and the edge region respectively. Wavelet transform can transform the advantages of the frequency domain even more. Recent work has begun to combine wavelet transform with CNN and apply it to classification, image compression and other fields. Liu et al. [35], propose a novel image mooring-removing method based on wavelet and two-branch neural network, which combines a dense convolution module and an extended convolution module for expanding the receptive field. Huang et al. [36] proposed the CNN method based on wavelet for face super resolution, which firstly predicts the wavelet coefficient and then reconstructs the high resolution image. In addition, the network also uses wavelet to predict the loss, which is used to converge during training. Yoo et al. [51] propose a wavelet correction transformation based on whitening and coloring transformations, allowing features to retain their structural information. Low-light enhancement in the wavelet domain is beneficial for noise suppression and key information recovery, but there is no work to apply wavelet transform to low light enhancement.

2.3. Transformer in low-level vision

Google [52] proposed Transformer, self-attention has a profound impact on a lot of later work. Transformer was originally used for NLP tasks and was inspired to propose vision Transformer (ViT) [53], which directly divides the image into fixed-size patches and then transforms them linearly. The advantages of Transformer have attracted attention, and many low-level vision tasks have begun to introduce Transformer recently. Wang et al. [33] proposed a U-shaped Transformer called UFormer, which focuses on reducing the amount of computation. Restormer is proposed by Zamir et al. [54] for image restoration of high-resolution images. They propose the multi-Dconv head transposed attention (MDTA) module, which aggregates local and non-local pixel interactions. In addition, gated-dconv feed-forward network (GDFN) is proposed, which performs controlled feature transformations. Chen et al. [55] proposed a new pre-training model IPT, which taps Transformer's capacity to the maximum extent through damaged images. IAT [31] uses digital image processing for least square adjustment, adjusts global image information through attention mechanism, and designs an ultra-lightweight real-time enhancement network with end-to-end Transformer structure.

2.4. Semantic segmentation

Semantic segmentation is one of the basic tasks of computer vision. Its goal is to make pixel-level classification prediction of images. Long et al. [56] proposed full convolutional neural networks (FCN). Zhao et al. [57] propose PSPNet to generate high-quality segmentation by building pyramidal pooling modules (PPM) to aggregate global and local information. Chen et al. [58] proposed DeepLabv3+, which uses depth-separable convolution for pyramidization of void space.

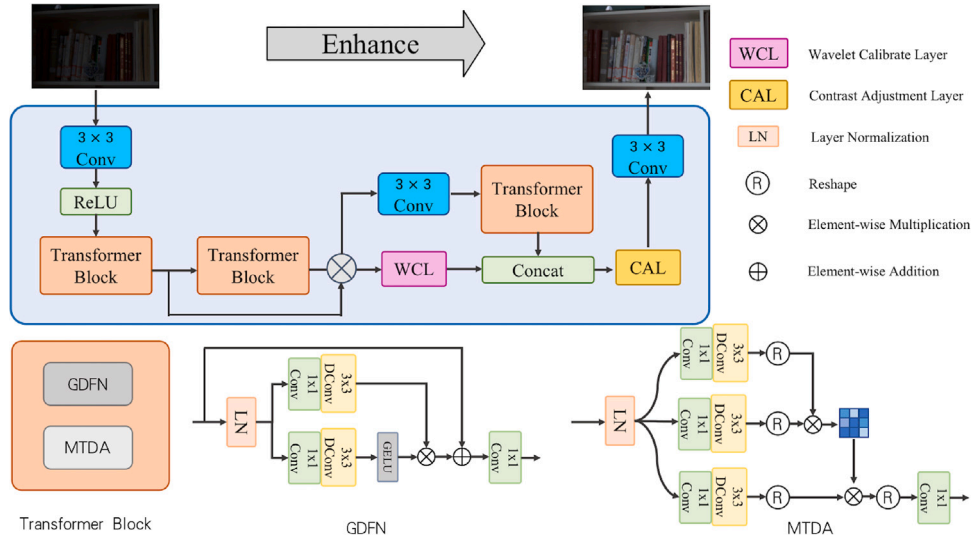


Fig. 2. Structure of WENet. WENet consists of Transformer block, wavelet calibrate layer (WCL) and contrast adjustment layer (CAL), which reduces the computing volume of the network as much as possible. WCL can regionally correct features in the wavelet domain, and CAL further improves the contrast of the image.

Fu et al. [59] proposed Dual Attention Network (DANet) to enhance feature representation in scene segmentation. They proposed a Position Attention Module (PAM) and a Channel Attention Module (CAM) to enrich context relevance. Yu et al. [60] proposed a real-time semantic segmentation network BiSeNet, spatial paths to store spatial information and generate high-resolution features, and context paths to increase the sensitivity field. Finally, a feature fusion module is used to get the prediction. Although the above models achieve good results on the baseline dataset, most of them are studied under normal light. The loss of information in a large area of the image under low light conditions will cause serious errors in the segmentation model, and the network is indeed robust to the dark environment. The purpose of this paper is to improve the performance of semantic segmentation by studying a low-light enhancement model.

3. Method

3.1. Overview of WENet

We hope to filter out the noise in the feature in the wavelet domain and recover the information lost in the image as much as possible. The structure of Wavelet-based enhancement network (WENet) is shown in Fig. 2. WENet consists of Transformer block, wavelet calibrate layer (WCL), and contrast adjustment layer (CAL). First, a low light image $I \in R^{H \times W \times 3}$ is taken as input, H and W represent the height and width of the image respectively, and $F_0 \in R^{H \times W \times 16}$ is obtained by learning shallow features through a convolution layer. After that, we construct two Transformer blocks to carry out deep learning on features and calculate a weight map M , obtaining $F_1 = F_0 \cdot M$, $F_1 \in R^{H \times W \times 16}$, then it is divided into two branches. On the one hand, use a 3×3 convolution layer to change the F_1 channel from 16 to 32, and then use a Transformer block for feature extraction. On the other hand, F_2 is fed into the wavelet calibrate layer (WCL) to correct the features in the wavelet domain. Then, we spliced the features of the two branches to get $F_3 \in R^{H \times W \times 48}$, and then fed a contrast adjustment layer (CAL) composed entirely of convolutional layers to improve contrast, and finally got the enhanced image.

Review transformer. In this section, we delve into the intricacies of the Transformer architecture. Specifically, within the Transformer block of our network, we adopt the Restormer structure [54]. Restormer has demonstrated superior performance across a variety of image restoration tasks by introducing several innovative key designs within its building blocks, enabling the capture of long-range pixel

interactions. It should be noted that the focus of our research in this paper is WCL and CAL. transformer block does not require too much attention, it can also be swin transformer, etc. Restormer improved several key modules so that it can capture remote dependencies while still being suitable for large images. Restormer proposed multi-dconv head transposed attention (MDTA) and gated-dconv feed-forward network (GDFN). The details are shown in Fig. 2. MDTA implicitly encodes global context information by calculating attention on channels, and uses depth-wise convolution to generate Q , K , and V between calculating self-attention map to emphasize local information. The process is described as follows

$$\hat{X} = W_p \cdot \hat{V} \cdot \text{Softmax}(\hat{K}, \hat{Q}/\alpha) \quad (2)$$

where X is the inputs, W_p represents the parameters of the convolutional layer, and α is the factor. GDFN adopts a gating mechanism and depth-wise convolution to obtain the activation state of pixels. This process is described as

$$\hat{X} = W_p^0 \text{Gating}(x) + X \quad (3)$$

$$\text{Gating}(X) = \Phi(W_d^1 W_p^1 (LN(X))) \cdot W_d^2 W_p^2 (LN(X)) \quad (4)$$

where W_p represents ordinary 1×1 convolution, W_d represents depth-wise convolution, LN represents layer normalization, and Φ represents activation function.

3.2. Wavelet calibrate layer

Compared to CNN, Transformer has a greater advantage in establishing remote dependencies and can capture more correlation of features. Therefore, we learned from the Transformer block in front of the wavelet calibrate layer (WCL). Wavelet transform is a novel image transform method. Its main advantages are the localization analysis of spatio-temporal frequency and the multi-scale division of signal by stretching translation operation. Through the transformation, the different frequency components in the image can be well separated. In this process, we first need to select the wavelet basis function to calculate the wavelet coefficients, and then decompose the image into multiple sub-bands, which respectively contain low-resolution coarse information and high-resolution texture. This process is described as

$$c_{j,k_1,k_2} = \frac{1}{2^j} \iint f(x_1, x_2) \psi \left(\frac{x_1 - k_1 2^j}{2^j}, \frac{x_2 - k_2 2^j}{2^j} \right) dx_1 dx_2 \quad (5)$$

where $f(x_1, x_2)$ is the original image, ψ is the wavelet basis function, j represents the scale, k_1 and k_2 represent the position respectively,

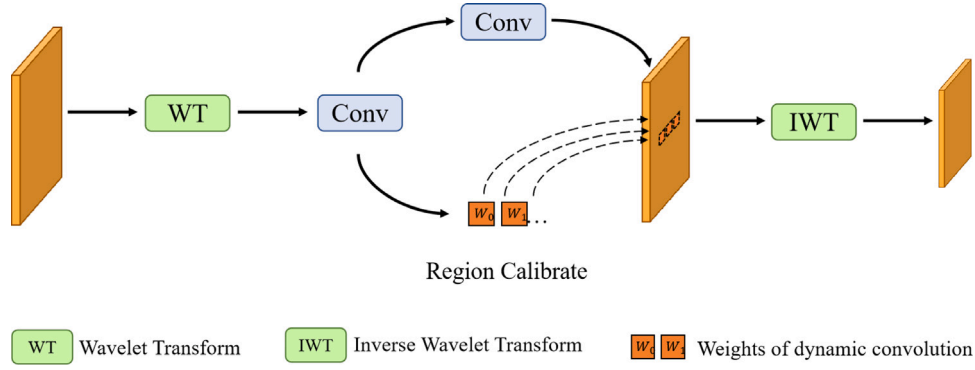


Fig. 3. Details of wavelet calibrate layer (WCL). We calibrate the region of features in the wavelet domain to recover more semantic information.

c_{j,k_1,k_2} is wavelet coefficients. Repeat the decomposition process to obtain a wavelet decomposition tree. Finally, we can reconstruct the original image by performing the reverse operation on the wavelet coefficients, this process is described as

$$f(x) = \sum_{j,k} c_{j,k} \psi_{j,k}(x) \quad (6)$$

where $\psi_{j,k}(x)$ is the wavelet basis function with scale j and position k .

We have observed that performing low-light enhancement in the wavelet domain is particularly effective for filtering out high-frequency noise while retrieving crucial low-frequency information. To facilitate this process, Wavelet Calibrate Layer (WCL) is designed to meticulously adjust features within the wavelet domain. The detailed workings of the WCL are illustrated in Fig. 3. First, the input feature $F_{in} \in R^{H \times W \times C}$ is transformed into a wavelet domain and sliced into F' by the wavelet transform. We use 2D fast wavelet transform (FWT) to decompose the image into a sequence of wavelet coefficients with different frequency components. The original input is converted into 48 subbands, and we connect all 48 channels together instead of dealing with the high and low bands separately, because it is difficult for us to accurately separate them.

After that, the features are corrected at the regional level. We define $S = \{s_0, s_1, s_2, \dots, s_{n-1}\}$ for the division of regions. Each region is convolved by a 3×3 depth-wise convolution, and the output for each region is

$$y_{i,j,o} = \sum_{c=0}^C x_{i,j,c} \cdot W_o^c \quad (7)$$

where y is the output result, o is the region, C is the number of channels, x is the current input F' , and W_o^c represents the filter parameter in region o at the time of channel c . After that, what we need to consider is how the filter chooses the area to process. We use learnable guided mask [61], which determines the distribution of filters in the spatial dimension. We use 3×3 convolution to generate mask features $F_{mask} \in R^{H \times W \times n}$ with n channels. for each position (i, j) , the mask M is

$$M_{i,j} = \operatorname{argmax}(F_{i,j}^0, F_{i,j}^1, F_{i,j}^2, \dots, F_{i,j}^{n-1}) \quad (8)$$

where argmax is the index of the maximum value. The value of M represents the index of the filter used at the corresponding position. After the region correction, we restore the feature to the spatial domain by inverse wavelet transformation to obtain $F_{out} \in R^{H \times W \times C}$. In addition, we construct wavelet transform and inverse wavelet transform by convolutional layer and deconvolution layer respectively, and load the prepared wavelet parameters in the two layers.

Within the wavelet domain, the frequency components are adeptly separated and subsequently reconstructed through a process of regional correction. The majority of noise present in low-light images is attributed to high-frequency components, whereas the essential semantic information is predominantly embedded in the low frequencies. By employing this approach, the enhanced images we produce are endowed with richer information content, thereby providing a significant advantage for subsequent tasks such as semantic segmentation.

3.3. Contrast adjustment layer

The origin image is in the spatial domain, and there will be a gap between spatial domain and wavelet domain. To solve this problem, after learning the features of the wavelet domain, we constructed a contrast adjustment layer (CAL) to finetune the features of wavelet domain learning, the details of CAL are shown in Fig. 4. We take the feature F_3 formed by the previous two branches as input, and the number of channels is dim . We first go through a 3×3 convolution by changing the number of channels to $2 * dim$. Further feature extraction is carried out by a convolutional layer, Batch Normalization and ReLU. Then, we adjust the features with a shift operation, which helps improve the contrast of the image. The multiplicative dimension and the additive dimension of the feature are adjusted by two branches respectively, the first convolution of each branch doubles the number of channels of the input feature, and the second convolution returns the number of channels of the feature to the same as the input.

According to the least squares method, we have adopted a simple shift operation. This process is described as

$$S(x) = F_m(x) * x + F_a(x) \quad (9)$$

where F_m is the filter branch of multiplication, and F_a is the filter branch of addition. This is similar to the least squares method, which is adjusted separately from the angle of multiplication and addition. The process of each branch is

$$F(x) = W_2(\Phi(W_1(x))) \quad (10)$$

where $W_i (i = 1, 2)$ represents the convolution layer parameters, Φ represents ReLU. Finally, we obtain a set of weight matrix through convolution and sigmoid function, and make residuals with the features of input CAL to get the output. CAL adjustment can prevent the feature from having a large offset in the wavelet domain, and shift operation can also improve the contrast of the enhanced feature.

3.4. Loss function and finetune model

We are not very complex in the design of the loss function, and only use the loss of normal low-light enhancement, including L1 loss and perception loss through pre-trained VGGNet. Our goal is to improve human visual experience and the performance of semantic segmentation task. WENet generates high quality images to improve the performance of segmentation model. We embed WENet into DeepLabv3+ for end-to-end training, and fine-tune WENet in the training process. In training, we only use the cross entropy loss for segmentor convergence.

4. Experiments

4.1. Datasets and implementation details

We used the LOL and LOLv2 dataset to verify the validity of our model, and the ACDC dataset to verify the performance improvement of

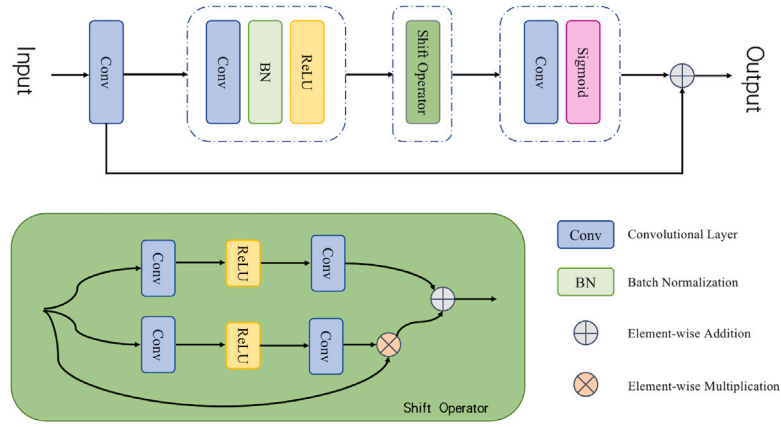


Fig. 4. Details of contrast adjustment layer (CAL). In this we adopt a shift operate to improve contrast.

Table 1

Comparison of low-light enhancement model in LOL dataset. The first best result and the second best result are marked in blue and green, respectively (see [62,63]).

Model	From	PSNR	SSIM	LPIPS	Parameters (M)
RetinexNet [27]	BMVC2018	16.77	0.562	0.474	0.84
MBLLEN [26]	BMVC2018	17.86	0.727	0.225	20.47
KIND [25]	MM2019	17.65	0.775	0.175	8.16
KIND++ [62]	IJCV2021	17.75	0.76	–	8.28
Zero-DCE [29]	CVPR2020	14.86	0.589	0.335	0.08
DSLR [64]	TMM2020	15.05	0.597	0.376	14.93
LIME [13]	TIP2016	16.75	0.445	0.395	–
IPT [55]	CVPR2021	16.27	0.504	–	115.63
EnlightenGAN [28]	TIP2021	17.48	0.677	0.322	8.64
Uformer [33]	CVPR2022	16.36	0.771	0.321	5.29
IAT [31]	BMVC2022	21.87	0.788	–	0.09
SCI [30]	CVPR2022	14.78	0.522	0.339	0.58
PairLIE [63]	CVPR2023	19.51	0.736	0.248	3.41
WENet (ours)	–	22.34	0.814	0.298	3.39

the low-light enhanced model on downstream semantic segmentation task. LOL contains 500 pairs of low-light/normal images, all sized to 400×600 , and has become a common dataset for low-light enhancement. LOLv2 contains 689 pairs of images for training and 100 pairs of images for testing. The MIT-Adobe FiveK dataset contains 5000 images, each of which is manually enhanced by five different experts. ACDC dataset can be used to train and test semantic segmentation models under unfavorable visual conditions. The ACDC dataset contains 4006 images distributed evenly across four common adverse weather: fog, night, rain, and snow, with high-quality pixel-level labeling for each image. In this paper, we only use images under night conditions.

Our devices are NVIDIA RTX 3090 GPU, Pytorch, Ubuntu18.04, etc. In the low-light enhancement experiments, we used SGD as the optimizer and set the initial learning rate and weight decay to 0.0001 and 0.0005, respectively. When training, we set batch_size to 8 and epoch to 500. In the dark semantic segmentation experiments, we adopt mmsegmentation to apply the most basic data augmentation methods such as random flipping and cropping. The optimizer uses SGD with an initial learning rate of 0.005 and weight decay of 0.0005. Set batch_size to 8 and iteration to 20 000.

When testing low-light enhancement, we used PSNR, SSIM, LPIPS, NIQE to verify the accuracy of the model. We used mIoU to evaluate the segmentation model. IoU represents the ratio of the intersection area and union area between prediction and ground truth, which is also defined as

$$IoU = \frac{TP}{TP + FP + FN} \quad (11)$$

where TP represents true positive, FP represents false positive, and FN represents false negative. $mIoU$ is the average value of the IoU for each class.

4.2. Comparison of low-light image enhancement

We compared WENet with other low-light enhancement models in the LOL dataset, and the results are shown in Table 1. We compared normal SOTA models such as RetinexNet, IAT, Kind, Zero-DCE, PairLIE, etc. Our WENet achieved the best results in both PSNR and SSIM, 22.34 and 0.814 respectively. The PSNR and SSIM of WENet are 7.56 and 0.292 higher than SCI, and 2.83 and 0.078 higher than PairLIE. In addition, our parameter is only 3.39M, which has little impact on the real-time performance of downstream visual tasks. MBLLEN, RetinexNet, EnlightenGAN and other models lag behind in PSNR and the number of parameters and have no advantage. Both Uformer and IPT are enhancement models based on Transformer, but they do not have high performance in image quality while paying a lot of calculations. Although the number of parameters of Zero-DCE is small, it is based on the model sum of zero-shot learning, and the PSNR is low. Although IAT is light in parameter, our model is 0.47 and 0.026 higher in PSNR and SSIM, respectively.

We have also compared ours to other low-light enhancement models on the LOLv2 dataset, as shown in Table 2. Our WENet has achieved superior results, outperforming other methods in low-light image enhancement. While DRBN secures the second-best performance, it underperforms notably on the LOL dataset. Although IPT shows a close similarity to our method in terms of SSIM, its substantial parameters—115.63M, renders it less suitable for real-time applications due to increased computational demands. Our model is 1.80 and 0.064 higher than Kind on PSNR and SSIM, and 2.72 and 0.054 higher than Uformer on PSNR and SSIM, respectively. Our performance is also leading in models such as MBLLEN and Zero-DCE. In the MIT-Adobe FiveK dataset, our model achieves advanced performance, as shown in Table 3. Our model has reached 25.23, 0.925 and 0.091 on PSNR, SSIM and LPIPS respectively, proving the high visual quality of image enhancement. At the same time, we did a generalization experiment in the MIT-Adobe FiveK dataset, and the results are shown in Table 3. We will test the MIT-Adobe FiveK dataset through the LOL-trained model. The results show that although the domains between the two datasets are very different, our model still has good enhancement capabilities, reaching 17.97 and 0.743 on PSNR and SSIM respectively. For the assessment of non-reference image quality metrics, we utilized the Natural Image Quality Evaluator (NIQE) on the MIT-Adobe FiveK dataset, with the outcomes presented in Table 4. The results align with expectations, as supervised learning models typically do not excel in non-reference metrics, which are designed to assess image quality without direct comparison to a reference image.

In addition, our model has a low value on LPIPS, which shows that the image quality is similar to human perception. We visualized the enhanced results of some models on LOL dataset, as shown in Fig. 5. Our model achieved a good visual experience, but in contrast,

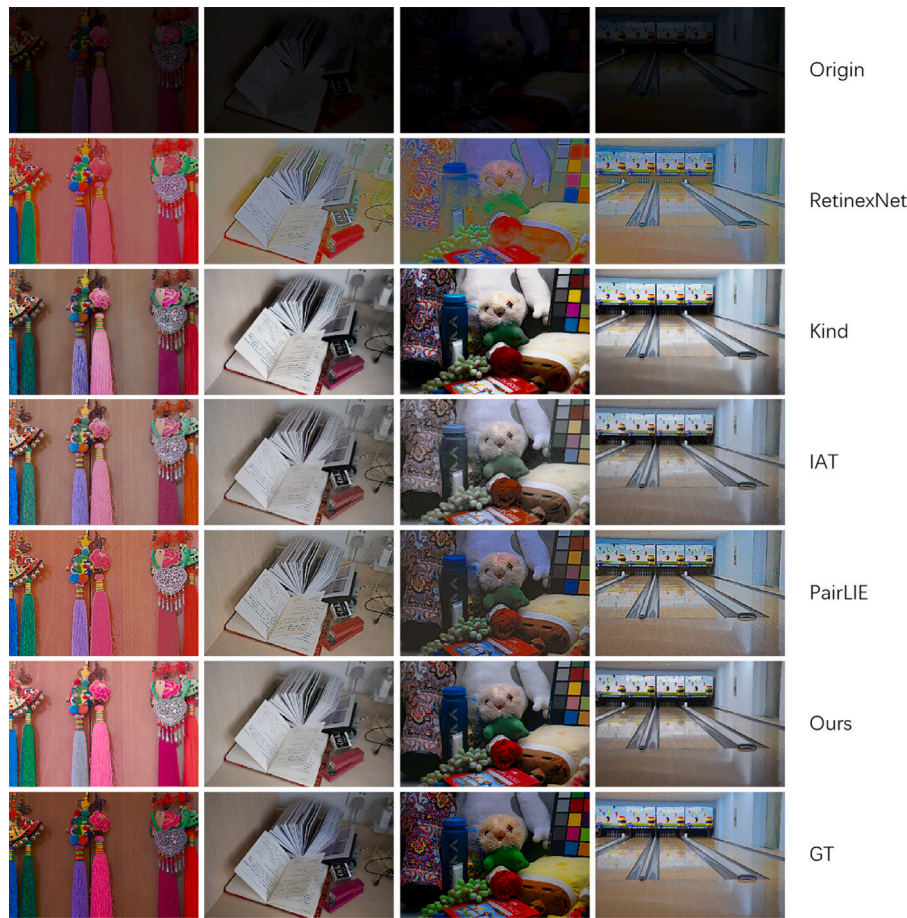


Fig. 5. Visualization of low-light enhancement model. Each column is a different image example, and each row is the prediction of models.

Table 2

Comparison of low-light enhancement model in LOLv2 dataset. The first best result and the second best result are marked in blue and green, respectively.

Model	From	PSNR	SSIM	LPIPS	Parameters (M)
RetinexNet [27]	BMVC2018	18.37	0.723	0.365	0.84
MBLLEN [26]	BMVC2018	18.00	0.715	–	20.47
KIND [25]	MM2019	19.74	0.761	0.375	8.16
DRBN [65]	CVPR2020	20.13	0.820	0.147	–
IPT [55]	CVPR2021	19.80	0.813	–	115.63
IAT [31]	BMVC2022	21.30	0.815	0.325	0.09
UFormer [33]	CVPR2022	18.82	0.771	0.347	5.29
WENet (ours)	–	21.54	0.825	0.305	3.39

Table 3

Comparison of low-light enhancement model in MIT-Adobe FiveK dataset. The first best result and the second best result are marked in blue and green, respectively.

Model	Venue	PSNR	SSIM	LPIPS
RetinexNet [27]	BMVC2018	12.51	0.671	0.254
Kind [25]	MM2019	14.54	0.741	0.150
Zero-DCE [29]	MM2019	13.20	0.709	0.165
DSLR [64]	TMM2020	16.63	0.782	0.153
EnlightenGAN [28]	TIP2021	13.26	0.745	0.143
UFormer [33]	CVPR2022	21.92	0.870	0.090
Restormer [54]	CVPR2022	24.92	0.911	0.058
SCI [30]	CVPR2022	20.44	0.893	–
LLFormer [66]	AAAI2023	25.75	0.923	0.045
WENet (ours)-LOL	–	17.97	0.743	0.341
WENet (ours)	–	25.23	0.925	0.091

Table 4

Comparison of NIQE in MIT-Adobe FiveK dataset. The best result are marked in blue.

Model	NIQE
RetinexNet [27]	4.52
Kind [25]	4.27
DRBN [65]	4.82
Zero-DCE [29]	4.10
WENet (ours)	4.61

RetinexNet showed serious color imbalance, Kind was a little excessive in contrast adjustment, and PairLIE's enhancement looked unnatural. WENet's visualization shows that the enhanced image is more natural and comfortable, and the contrast looks more comfortable. In addition, We visualized the enhanced results on LOLv2 dataset in Fig. 6, our model still have pleasant visual experience.

4.3. Comparison of dark semantic segmentation

We use different low-light enhancement models for semantic segmentation, firstly we load the model weight trained by the LOL dataset. and fine-tune it through end-to-end training to make the segmentation robust to the scene. Here the segmentation model is all using DeepLabv3+ without any modification. The experimental results are shown in Table 5. It should be noted that we tested it under 256×256 , and the original shape is 1920×1080 . The purpose of the experiment in this section is to test the effectiveness of the enhancement model for downstream tasks. We found that Zero-DCE and DeepLPF reached 61.9% in mIoU, an increase of only 0.2% compared with baseline, while IAT showed a decline in segmentation performance. Our WENet

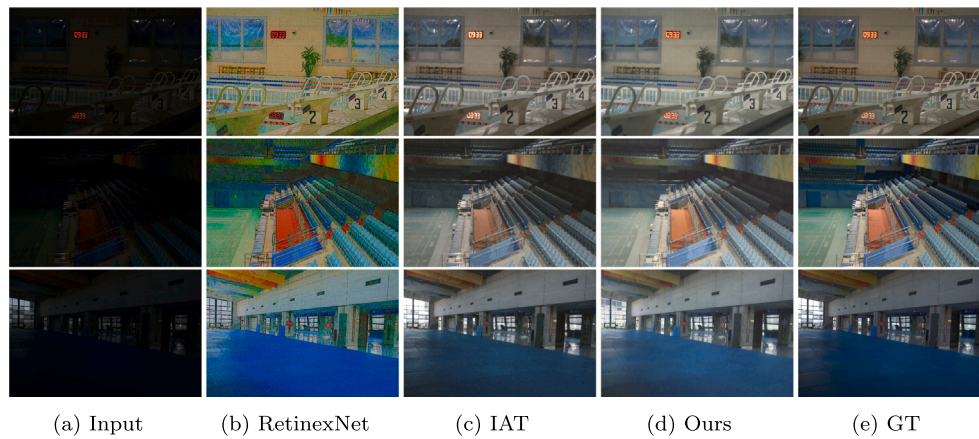


Fig. 6. Visualization of different low-light image enhancement methods on LOLv2 dataset. Each row is a different image comparison.

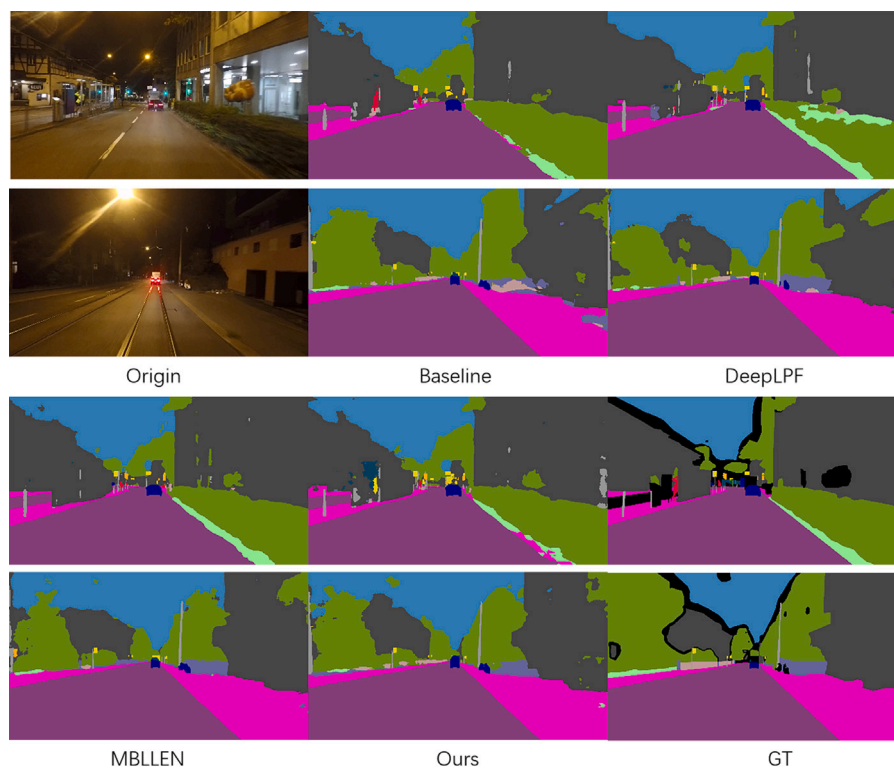


Fig. 7. Visualization of night semantic segmentation by different low-light enhancement models. In the figure is the segmentation effect of two images under different models, and our WENet has a good improvement in error prediction.

reached 62.4% on mIoU, an increase of 0.7%, and we only brought less than 0.3 s of time. This proves that our model improves the performance of segmentation task and does not have much impact on real-time. Fig. 7 shows the night semantic segmentation effect under different low-light enhancement models. It can be seen that our WENet can effectively reduce error prediction when used for segmentation after enhancing the low-light image.

4.4. Ablation studies

To verify the each part of WENet is effective, we get ablation studies, and the results are shown in Table 6. After adopting the wavelet calibrate layer, the PSNR is from 21.47 to 22.06 and SSIM is from 0.757 to 0.770, which shows that WENet is effective in wavelet domain for feature correction and can effectively filter out high-frequency noise. After applying the contrast adjustment layer, PSNR is from 22.06

Table 5

Comparison of semantic segmentation. We use different low-light enhancement methods for the training of semantic segmentation models, and the resolution of test set is 256×256 . The best result is marked by blue.

Segmentor	Model	mIoU (%)	Time (ms)	Parameters (M)
DeepLabv3+	Baseline	61.7	8.3	43.59
	Zero-DCE [29]	61.9	11.2	43.67
	DeepLPF [67]	61.9	26.9	45.55
	IAT [31]	61.5	9.3	43.68
	Ours	62.4	13.54	46.98

to 22.34 and SSIM is from 0.770 to 0.814, indicating that CAL can further adjust the features and balance the contrast. After adopting shift operator in CAL, PSNR is from 22.18 to 22.34 and SSIM ranges from 0.811 to 0.814, illustrating its effect. Compared to baseline, our model improved by 0.87 and 0.057 in PSNR and SSIM, respectively.

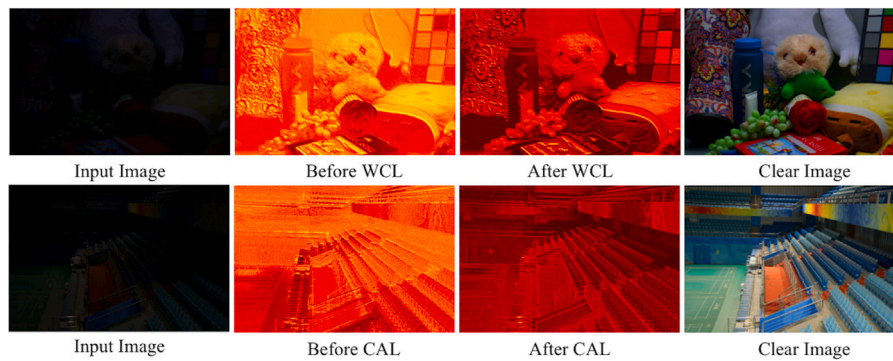


Fig. 8. Feature Visualization of WCL and CAL. WCL effectively restores the details of low-light images, while CAL balances the contrast of features.

Table 6

Results of ablation studies for WENet. The best result are marked in blue.

WCL	CAL		PSNR	SSIM
	w/o shift operator	w shift operator		
✓			21.47	0.757
✓	✓		22.06	0.770
✓			22.18	0.811
✓		✓	22.34	0.814

In addition, we visualized the features before and after WCL and CAL respectively, as shown in Fig. 8. Qualitative experiments showed that our WCL effectively restores the information of different frequency bands in the image through feature calibration in wavelet transform, while CAL reduces the gap between different domains and balances the contrast of the image through shift operation.

5. Conclusion

In the realm of the wavelet domain, we introduce the Wavelet-based Enhancement Network (WENet), an innovative framework that unifies convolutional layers with Transformer blocks. We have proposed a Wavelet Calibrate Layer (WCL) and a Contrast Adjustment Layer (CAL) to effectively filter out noise and enhance image contrast. Our WENet has demonstrated superior performance on the LOL and LOLv2 datasets. Furthermore, by jointly training WENet with a segmentation model in the dark scenes within the ACDC dataset, we have achieved commendable segmentation results. This integrated approach fosters robustness in low-light conditions, which is crucial for accurate scene interpretation under challenging lighting.

CRedit authorship contribution statement

XiaoPeng Hu: Software Programming, Formal analysis, Writing – original draft. **Kang Liu:** Visualization, Investigation, Supervision. **Xiangchen Yin:** Conceptualization, Validation. **Xin Gao:** Supervision, Project administration. **Pingsheng Jiang:** Visualization. **Xu Qian:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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